

Last Time:

• $\mathbb{R}^n = \{ \underbrace{(x_1, \dots, x_n)}_{\text{n-tuple of real numbers}} : x_i \in \mathbb{R} \text{ for every } \underbrace{\forall i}_{\text{\#s}} \}$

• $\mathbb{R}^n$ has two operations $(X \times Y = \{(x, y) : x \in X, y \in Y\})$

◇ Addition $\overset{\text{fcn}}{\text{function}} \downarrow \uparrow \overset{\text{domain}}{\text{domain}} \quad + : \mathbb{R}^n \times \mathbb{R}^n \xrightarrow{\quad \quad \quad} \overset{\text{codomain}}{\text{codomain}} \mathbb{R}^n$
 $((x_1, \dots, x_n), (y_1, \dots, y_n)) \mapsto (x_1 + y_1, \dots, x_n + y_n)$

means left element is mapped to right element

◇ Scalar multiplication: $\cdot : \overset{\text{set of scalars}}{\mathbb{R}} \times \mathbb{R}^n \xrightarrow{\quad \quad \quad} \mathbb{R}^n$
 $(c, (x_1, \dots, x_n)) \mapsto (cx_1, \dots, cx_n)$

• $\mathbb{R}^n$ with $+$ & $\cdot$ satisfy:

(VS 1) $\forall v, w \in \mathbb{R}^n, v + w = w + v$

(VS 2) $\forall u, v, w \in \mathbb{R}^n, (u + v) + w = u + (v + w)$

(VS 3) There $\exists$ exists $0_{\mathbb{R}^n} \in \mathbb{R}^n$ (or simply called 0)

such $\overset{\text{s.t.}}{\text{that}} \forall v \in \mathbb{R}^n, v + 0_{\mathbb{R}^n} = v = 0_{\mathbb{R}^n} + v$

(VS 4) $\forall v \in \mathbb{R}^n, \exists w \in \mathbb{R}^n$ s.t. $v + w = 0_{\mathbb{R}^n}$
 $-v := (-1) \cdot v$

(VS 5) $\forall v \in \mathbb{R}^n, 1 \cdot v = v$

(VS 6) $\forall a, b \in \mathbb{R}$ and $\forall v \in \mathbb{R}^n, (ab) \cdot v = a \cdot (b \cdot v)$

(VS 7) $\forall c \in \mathbb{R}$ and $\forall v, w \in \mathbb{R}^n, c(v + w) = cv + cw$

(VS 8) $\forall a, b \in \mathbb{R}$ and $\forall v \in \mathbb{R}^n, (a + b)v = av + bv$

Today: (1) Finish §1.1 $\mathbb{R}^n$ as vector space

(2) Start §1.2 Vector Spaces and Subspaces

Notation 1.7: In this course, we need to view

$\mathbb{R}^n$ as a "vector space" of "column vectors".

(mathematician's viewpoint)

n rows,
single column,

each $x_i \in \mathbb{R} \ \forall i$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$:= (x_1, \dots, x_n)$$

n -tuple of $\mathbb{R}$ #'s

col vec
column vector of size n (over $\mathbb{R}$)

$\Downarrow$

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{R} \ \forall i \right\} = \{ (x_1, \dots, x_n) : x_i \in \mathbb{R} \ \forall i \}$$

and

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{bmatrix} \quad \forall \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}, \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \in \mathbb{R}^n,$$

Def
1.4

$$c \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} ca_1 \\ \vdots \\ ca_n \end{bmatrix} \quad \forall \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n \ \& \ \forall c \in \mathbb{R}$$

§1.2 Vector Spaces and Subspaces

Motivation: In §1.1, we saw the properties of $+$ and $\cdot$ as operations on $\mathbb{R}^n$ (Prop 1.6 (VS1) – (VS8)).

As said before, want to define

- Vector: ^{elts}elements that can be added & scalar-multiplied ^{mult}
- Vector Space: suitable set of vectors w/ $+$ & $\cdot$.

Q: What else behaves like $\mathbb{R}^n$?

A: Functions ^{fns} $f, g: \mathbb{R} \rightarrow \mathbb{R}$ can be added & scalar-multiplied.

$$\begin{array}{ccc} \underline{f+g}: \mathbb{R} \rightarrow \mathbb{R}, & & \underline{c \cdot f}: \mathbb{R} \rightarrow \mathbb{R} \\ \uparrow & x \mapsto f(x) + g(x) & \uparrow \quad x \mapsto c f(x) \end{array}$$

both are fns! They "look like vectors".

Example

Ex 1.8: Let X be a set. Define ^{Def}

$\mathcal{F}(X, \mathbb{R}) :=$ set of all fns from X to $\mathbb{R}$.

For instance, $h: \mathbb{R} \rightarrow \mathbb{R}$ is in $\mathcal{F}(\mathbb{R}, \mathbb{R})$.
 $x \mapsto |x|$

^{addition}
Def $+: \mathcal{F}(X, \mathbb{R}) \times \mathcal{F}(X, \mathbb{R}) \rightarrow \mathcal{F}(X, \mathbb{R})$

$$(f, g) \mapsto \left(\begin{array}{l} f+g: X \rightarrow \mathbb{R} \\ x \mapsto f(x) + g(x) \end{array} \right)$$

^{scalar-mult}

$\cdot: \mathbb{R} \times \mathcal{F}(X, \mathbb{R}) \rightarrow \mathcal{F}(X, \mathbb{R})$

$$(c, f) \mapsto \left(\begin{array}{l} cf: X \rightarrow \mathbb{R} \\ x \mapsto cf(x) \end{array} \right)$$

$\Downarrow$

Claim: $\mathcal{F}(X, \mathbb{R})$ w/ $+$ & $\cdot$ satisfy (VS1) – (VS8)

from Prop 1.6 when replacing $\mathbb{R}^n$ by $\mathcal{F}(X, \mathbb{R})$.

(And in (VS 3), $0_{\mathcal{F}(X, \mathbb{R})} \stackrel{\text{if } X \neq \emptyset}{=} 0: X \rightarrow \mathbb{R}$,
 $x \mapsto 0$
unique
 in (VS 4), $w = (-1) \cdot v: X \rightarrow \mathbb{R}$
 $x \mapsto (-1)v(x)$ } Exercise

Proof

Pf Idea: $+$ & $\cdot$ are defined via outputs of fns,
 which live in $\mathbb{R}$. And $\mathbb{R}$ satisfies (VS 1) – (VS 8),
 so $\mathcal{F}(X, \mathbb{R})$ satisfies (VS 1) – (VS 8) as well.

Actual Pf of (VS 1) (rest are Exercises):

For $\forall$ every $f, g \in \mathcal{F}(X, \mathbb{R})$, $\forall x \in X$,

$(f+g)(x) = f(x) + g(x) \stackrel{\text{VS 1 for } \mathbb{R}}{=} g(x) + f(x) = (g+f)(x)$, so

$f+g = g+f$. □

Conclusion: (VS 1) – (VS 8) from Prop 1.6 (in §1.1)

are rules that determine when $+$ & $\cdot$ "behave like"
 $+$ & $\cdot$ from $\mathbb{R}^n$. (result of struggles of mathematicians)

Definition

Def 1.9: A vector space V (over $\mathbb{R}$) is a set

w/ addition $+: V \times V \rightarrow V$ and scalar-mult

$\cdot: \mathbb{R} \times V \rightarrow V$ satisfying (VS 1) – (VS 8) are called
axioms of v.s.

(VS 1) $\forall v, w \in V$, $v + w = w + v$

$$(VS 2) \forall u, v, w \in V, (u+v)+w = u+(v+w)$$

$$(VS 3) \text{ There } \exists \text{ exists } \underline{0_V} \in V \text{ such that } \forall v \in V, \\ v + 0_V = v = 0_V + v \text{ called zero-vector}$$

$$(VS 4) \forall v \in V, \exists \underline{w} \in V \text{ s.t. } v + w = 0_V \text{ called additive inverse of } v$$

$$(VS 5) \forall v \in V, 1 \cdot v = v$$

$$(VS 6) \forall a, b \in \mathbb{R} \text{ and } \forall v \in V, (ab) \cdot v = a \cdot (b \cdot v)$$

$$(VS 7) \forall c \in \mathbb{R} \text{ and } \forall v, w \in V, c(v+w) = cv + cw$$

$$(VS 8) \forall a, b \in \mathbb{R} \text{ and } \forall v \in V, (a+b)v = av + bv$$

Elements of V are called vectors (in V).

Remark

Rmk 1.10:

$$(a) \exists \text{ 1-1 correspondence } \mathbb{R}^n \xleftrightarrow{\quad} \mathcal{F}(\{1, \dots, n\}, \mathbb{R}) \\ (a_1, \dots, a_n) \mapsto \left(a: \{1, \dots, n\} \rightarrow \mathbb{R} \right. \\ \left. \begin{matrix} i & \mapsto & a_i \end{matrix} \right) \\ (f(1), \dots, f(n)) \leftarrow (f: \{1, \dots, n\} \rightarrow \mathbb{R})$$

Pf: Exercise

□

(b) $\mathbb{R}^n$ is a v.s. by Prop 1.6. So $\mathbb{R}^n$ is a v.s. of column ^{col} vecs of size n (also a v.s. of n -tuples of real ^{$\mathbb{R}$} ^{#s} numbers).

(c) $\mathcal{F}(X, \mathbb{R})$ is a v.s. of fns $X \rightarrow \mathbb{R}$ by Ex 1.8.

↑ generalizes $\mathbb{R}^n$ by (a)

Note: NO NEED TO MEMORIZE Statement #s (including axiom #s), as they depend on which books/notes. Those #s are NOT important! ^(waste)_(of time)

What's IMPORTANT is the CONTENT.

Ex 1.11: Let $\mathbb{R}^\infty := \mathcal{F}(\underbrace{\mathbb{Z}_{>0}}_{\{1,2,3,\dots\}}, \mathbb{R})$ be a v.s. (by Ex 1.8). Then just like Rmk 1.10(a), $\forall a \in \mathbb{R}^\infty$ can be thought of as a ^{seq} sequence $a(1), a(2), \dots$. Thus, seqs of $\mathbb{R}$ #s "are" vecs in $\mathbb{R}^\infty$.

Ex 1.12: An $m \times n$ ($m, n \in \mathbb{Z}_{>0}$) $(\mathbb{R}-)$ matrix is a rectangular array of $\mathbb{R}$ #s w/ m rows and n cols:

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix}$$

$A_{ij} = (A)_{ij}$ means the (i, j) -entry of A .

example,
 $B = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 4 \end{bmatrix}$
 $\uparrow$
 $M_{2 \times 3}(\mathbb{R})$
 $B_{2,2} = 3$

The set $M_{m \times n}(\mathbb{R})$ of all $m \times n$ matrices w/

◇ $A+B$ def by $(A+B)_{ij} := A_{ij} + B_{ij}$,

◇ cA def by $(cA)_{ij} := c(A_{ij})$,

is a v.s.

Pf: Exercise (Similar to Pf of Ex 1.8.) $\square$

Rmk 1.13:

(a) $\mathbb{R}^n = M_{n \times 1}(\mathbb{R})$ by def. $\left(\begin{array}{l} \text{col vec of} \\ \text{size } n \end{array} = \begin{array}{l} n \times 1 \\ \text{matrix} \end{array} \right)$

(b) A row vec of size n is $[a_1 \dots a_n] \in M_{1 \times n}(\mathbb{R})$.

For example, $[1 \ 3 \ 5] \in M_{1 \times 3}(\mathbb{R})$ v.s. of row
vecs of size n

$$\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \stackrel{\#}{=} (1, 3, 5) \in \mathbb{R}^3 = M_{3 \times 1}(\mathbb{R})$$

(c) $M_n(\mathbb{R}) := M_{n \times n}(\mathbb{R})$ v.s. of square matrices
of size n .

Ex 1.14: Consider $\mathbb{R}^2$ w/ usual scalar-mult.

and new addition $+': \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2$.

$$(a, b), (c, d) \mapsto (a+c, 0)$$

Then (VS3) fails for $v = (1, 1)$, because ^{b/c}

$\forall (c, d) \in \mathbb{R}^2, (1, 1) +'(c, d) = (1+c, 0) \neq (1, 1)$, so

^{not $\exists$} $\nexists 0_{\mathbb{R}^2}$ for $(1, 1) \in \mathbb{R}^2$ w/ $\textcircled{+}'$ & .

bad operation for $\mathbb{R}^2$ as v.s.

Properties of V.S.

Motivation: For $\mathbb{R}^n$ & $\mathcal{F}(X, \mathbb{R})$, 0 from (VS3)

and additive inverse from (VS4) are unique.

Also, $a+c = b+c \Rightarrow a=b$ on those two Exs.

Q: True for general v.s.?

⇓ Yes!

Lemma

Lem 1.15 (Cancellation Law): (obvious looking, but)
(not an axiom)

Let x, y, z be vecs in a v.s. V . If $x+z = y+z$,
then $x = y$. no need to cite
↓

Pf: Let w be an additive inverse of z (by VS4).

Then $x+z = y+z \xRightarrow{\text{implies}} (x+z)+w = (y+z)+w$

$\xRightarrow{\text{VS2}} x + \underbrace{(z+w)}_{0_V} = y + \underbrace{(z+w)}_{0_V} \xRightarrow{\text{VS4}} x + 0_V = y + 0_V$

$\xRightarrow{\text{VS3}} x = y$.

□